

LeadDesk Mini - Project & Deployment Guide

Complete Reference Manual • Architecture • User Guide • MongoDB Cloud • Live Hosting

1. Executive Summary & Capabilities

LeadDesk-Mini is a complete, production-grade Lead Management CRM system engineered for agile sales and customer acquisition teams. It bridges public inquiry capture with an admin workspace featuring real-time stage progression, internal collaboration notes, and search-and-filter capabilities.

Component	Key Technologies	Main Functionality
Frontend	React 19, Vite, Vanilla CSS	Landing Page, Public Demo Request, Kanban Board, Data Table, Internal Notes Drawer, CSV Export.
Backend	Node.js, Express.js (ES Modules)	RESTful API, JWT Authentication, Lead CRUD, Stage Updates, Query Filters, Route Protection.
Database	MongoDB Atlas (Cloud AWS)	Managed NoSQL document storage for Leads and Admin user credentials.

2. Step-by-Step User Guide

A. For Public Visitors & Inbound Prospects

- Navigate to the homepage at `http://localhost:5173/` (or your live public domain).
- Scroll to the **Request Demo / Contact** section.
- Fill in the form fields:
 - Full Name** (Required)
 - Work Email** (Required, validated format)
 - Phone Number** (Required)
 - Company Name** (Optional)
 - Estimated Budget** (Dropdown selection)
 - Project Message** (Optional)
- Click **Submit Request**. The inquiry is ingested and saved into the database instantly.

B. For Sales Representatives & Admin Teams

- Visit `/login` and authenticate using your registered Admin credentials.
- Inside the **Admin Dashboard (/admin)**:
 - Kanban Pipeline View**: Drag or transition leads across stages: *New* → *Contacted* → *Qualified* → *Lost*.

- **Table View:** Instant search by name or company, and filter by priority (*Urgent, High, Medium, Low*).
- **Lead Details & Internal Notes:** Click any lead to inspect details and append timestamped team notes.
- **CSV Export:** Click *Export CSV* to generate spreadsheet files for reporting.

3. MongoDB Atlas Database Details & Structure

MongoDB Atlas Cloud Connection

Cluster: Cluster0 (cluster0.dzrc2es.mongodb.net)

Database Name: leaddesk

DB User: jyotikumariroy204_db_user

Web Portal URL: <https://cloud.mongodb.com>

Document Schema Definitions

1. Leads Collection (leads):

```
{
  "_id": ObjectId("..."),
  "fullName": "Sarah Jenkins",
  "email": "sarah@acmecloud.com",
  "phone": "+1 555-0199",
  "company": "Acme Cloud Solutions",
  "budget": "$10k - $25k",
  "priority": "Urgent",    // Low | Medium | High | Urgent
  "status": "New",        // New | Contacted | Qualified | Lost
  "source": "Website Form",
  "notes": [
    {
      "text": "Initial discovery call scheduled for 3 PM.",
      "author": "Admin",
      "createdAt": ISODate("2026-09-16T15:30:00Z")
    }
  ],
  "createdAt": ISODate("..."),
  "updatedAt": ISODate("...")
}
```

2. Users Collection (users):

```
{
  "_id": ObjectId("..."),
  "email": "admin@leaddesk.com",
  "password": "$2a$10$hashedPasswordString...",
  "role": "Admin",
  "createdAt": ISODate("...")
}
```

Important: Atlas Network Access Setting

To enable cloud deployments (Render/Vercel) to communicate with MongoDB Atlas:

1. Log into **cloud.mongodb.com** → Click **Network Access**.
2. Ensure **0.0.0.0/0** (Allow Access from Anywhere) is added to the IP Access List.

4. Cloud Deployment & Sharing Guide

Step 1: Backend Deployment (Render.com)

1. Push repository to GitHub.
2. Go to **render.com** → **New +** → **Web Service**.
3. Set **Root Directory** to `server`, **Build Command** to `npm install`, and **Start Command** to `npm start`.
4. Add Environment Variables:
 - `PORT` = `5000`
 - `MONGODB_URI` = (Your Atlas Connection String)
 - `JWT_SECRET` = (Your secret key)

Step 2: Frontend Deployment (Vercel.com)

1. Go to **vercel.com** → Import your GitHub repo.
2. Set **Root Directory** to `client`.
3. Add Environment Variable:
 - `VITE_API_URL` = `https://your-backend-service.onrender.com`
4. Click **Deploy**. You will receive a custom public link (e.g. `https://leaddesk.vercel.app`).

Custom Domain Setup (Optional)

In Vercel → Project Settings → Domains → Enter your domain (e.g. `www.mycompany.com`) → Add the generated CNAME record in your domain registrar DNS settings.